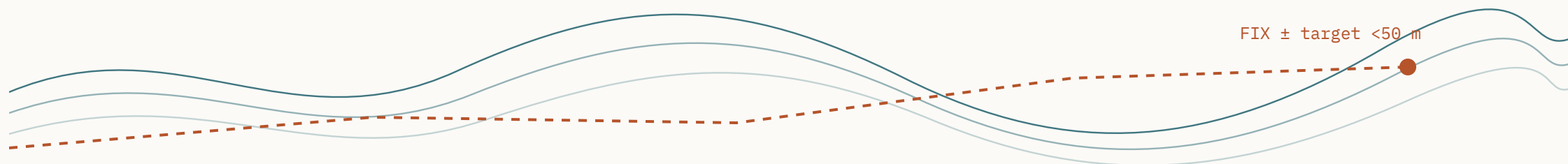


VOID-NAV: Navigating Without the Sky

A diamond quantum magnetometer that locates a platform against the Earth's own magnetic fingerprint, when GPS is jammed, spoofed or unavailable.

MAGNETIC ANOMALY CONTOURS · ESTIMATED TRACK



TEAM

Team VOID

DELIVERABLE

Working hardware prototype + fusion software

Modern forces fight in a contested electromagnetic spectrum. GPS is the first thing to go.

$\sim 10^{-16}$ W

Typical GNSS signal power at the receiver. A low-cost jammer overwhelms it over many kilometres.

~ 1.8 km

Indicative position error per hour for a tactical-grade inertial unit left uncorrected.

0

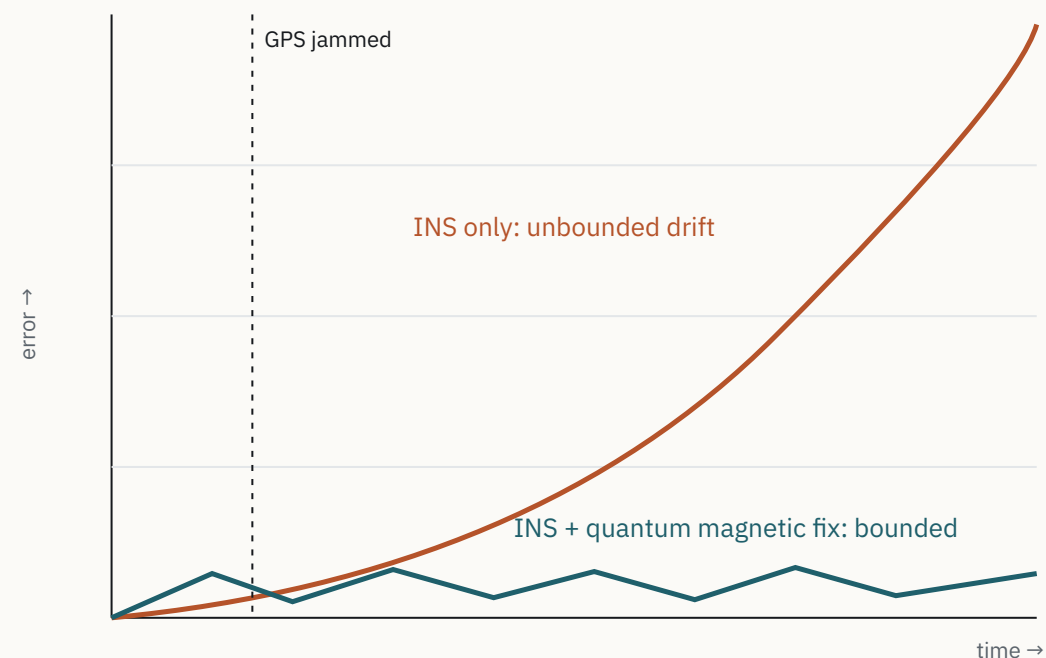
Signals that reach a submerged submarine: it navigates blind for weeks.

Passive

What any fix must be: an emitting sensor gives away the platform's position.

Figures are indicative, drawn from open literature; exact values vary by receiver and IMU class.

POSITION ERROR OVER TIME (ILLUSTRATIVE)



We scored all four defence use cases. UC-085 is the one we can prove on a bench.

USE CASE	DEFENCE URGENCY	QUANTUM ADVANTAGE TODAY	HARDWARE DEMO FEASIBLE	INDIGENOUS BUILD	SCORE
UC-084 · QKD secure comms	●●●●●	●●●●○	●●○○○	●●●○○	15
UC-085 · GPS-denied navigation	●●●●●	●●●●●	●●●●●	●●●●○	19
UC-086 · Radar/sonar QML	●●●●○	●●○○○	●●○○○	●●●○○	11
UC-087 · Logistics optimisation	●●●○○	●●○○○	●○○○○	●●●●○	10

Sensing is ready now

Quantum sensing is the most mature quantum pillar; it works at room temperature, not in a cryostat.

Mission-aligned

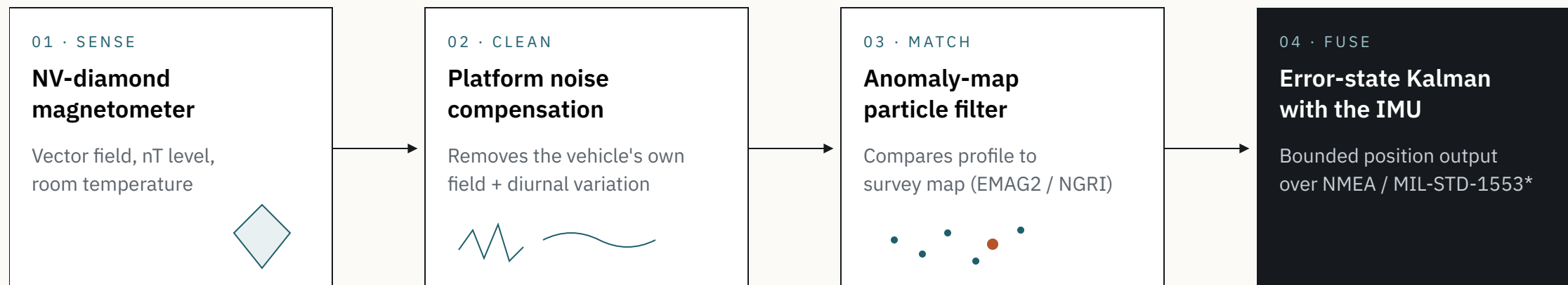
Directly supports the National Quantum Mission's sensing & metrology hub.

Visible proof

Judges watch a live resonance dip on screen; no simulation required.

The Earth's crust is a map that cannot be jammed.

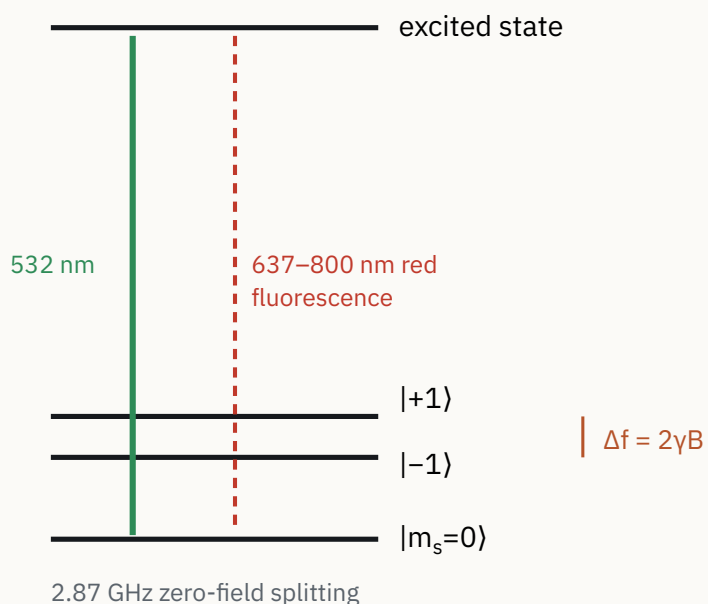
Rocks carry a unique, stable magnetic signature. A sensitive enough magnetometer reads it like a barcode; matching the reading to a survey map gives position, with no transmission.



Fully passive: nothing is emitted, so there is nothing to jam, spoof or detect.

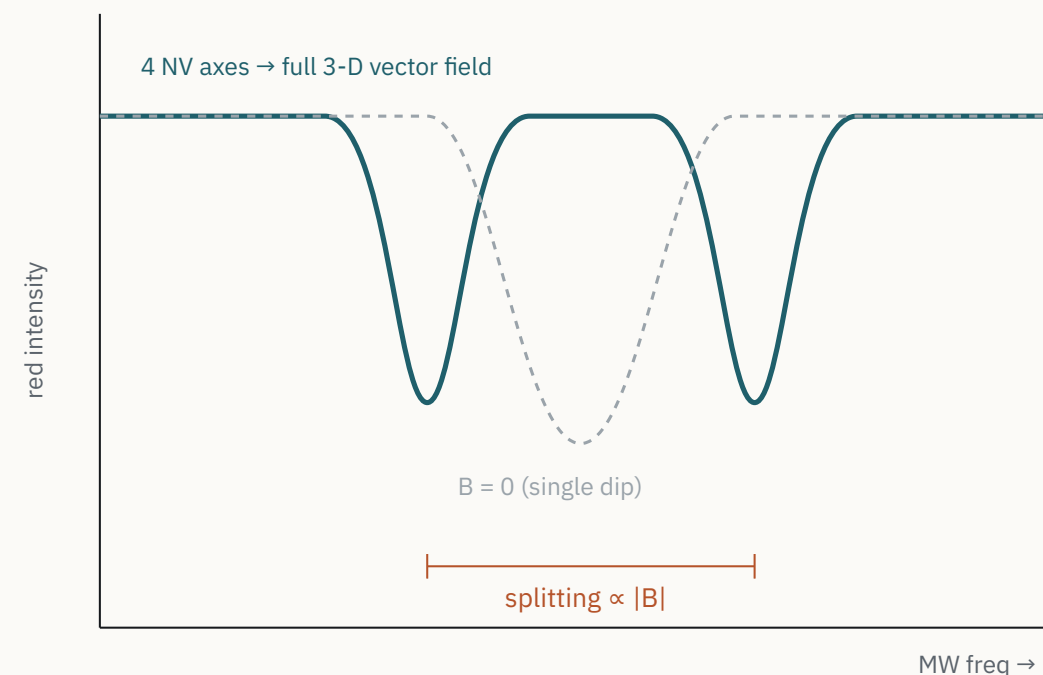
* 1553 interface is a Phase-2 target; prototype outputs NMEA over serial.

A single defect in diamond turns a magnetic field into a colour of light we can measure.

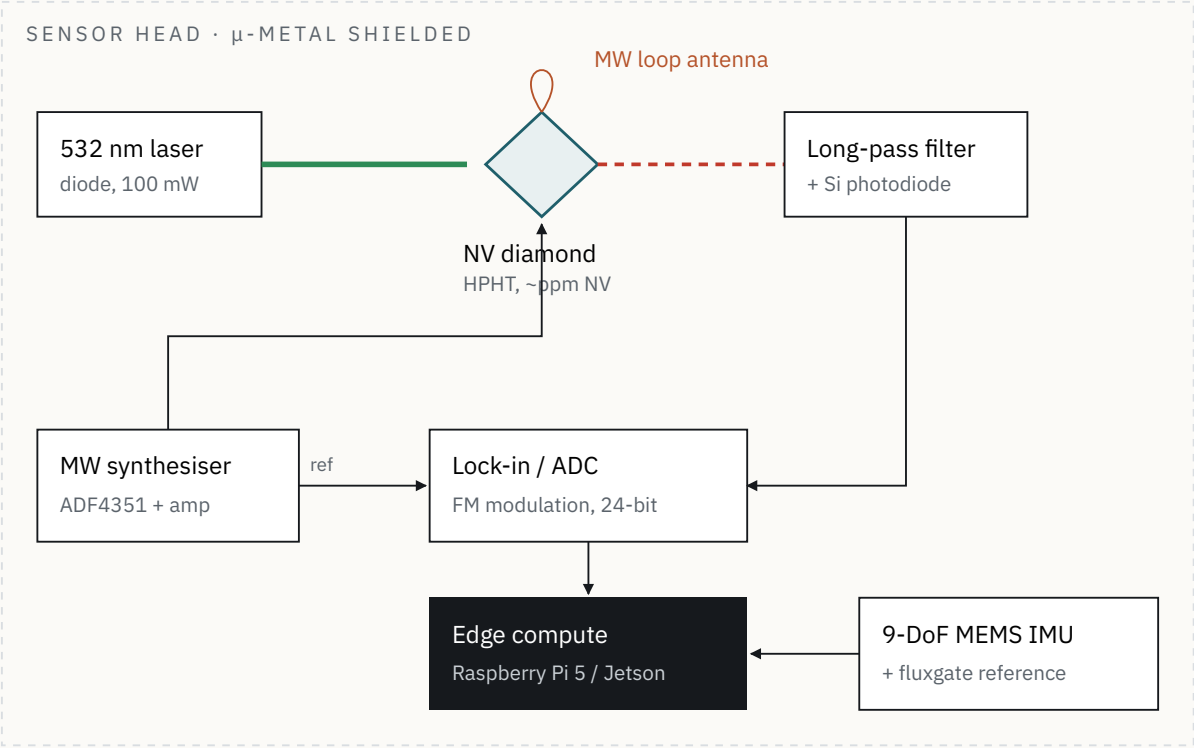
NV⁻ CENTRE · SPIN TRIPLET

$\gamma \approx 28 \text{ GHz/T}$: field shifts the microwave resonance

OPTICALLY DETECTED MAGNETIC RESONANCE



A shoebox-sized quantum sensor built from components we can source in India.



SUBSYSTEM	EST. COST (₹)
NV-diamond sample	25,000
532 nm laser + optics	15,000
MW synthesiser + amplifier	12,000
Photodiode, lock-in, ADC	8,000
Compute, IMU, shielding	15,000
Prototype total	≈ 75,000

Budgetary estimates. Commercial quantum navigation units cost orders of magnitude more.

- ROOM TEMPERATURE
- NO CRYOGENICS
- NO VACUUM
- COTS PARTS

Four contributions that go beyond a lab magnetometer.

N - 01

Vector, not scalar, map matching

The four NV axes give the full 3-D field. We match gradient direction as well as magnitude, which removes the ambiguity that limits scalar magnetic navigation over flat terrain.

N - 03

Confidence-aware fusion

The filter weights a quantum fix by local map "informativeness". Over magnetically bland regions it trusts the IMU; over rich anomalies it snaps the track back.

N - 02

Learned self-field compensation

A small on-device model learns the host vehicle's own magnetic signature (engines, current loops) during a short calibration run, then subtracts it in real time.

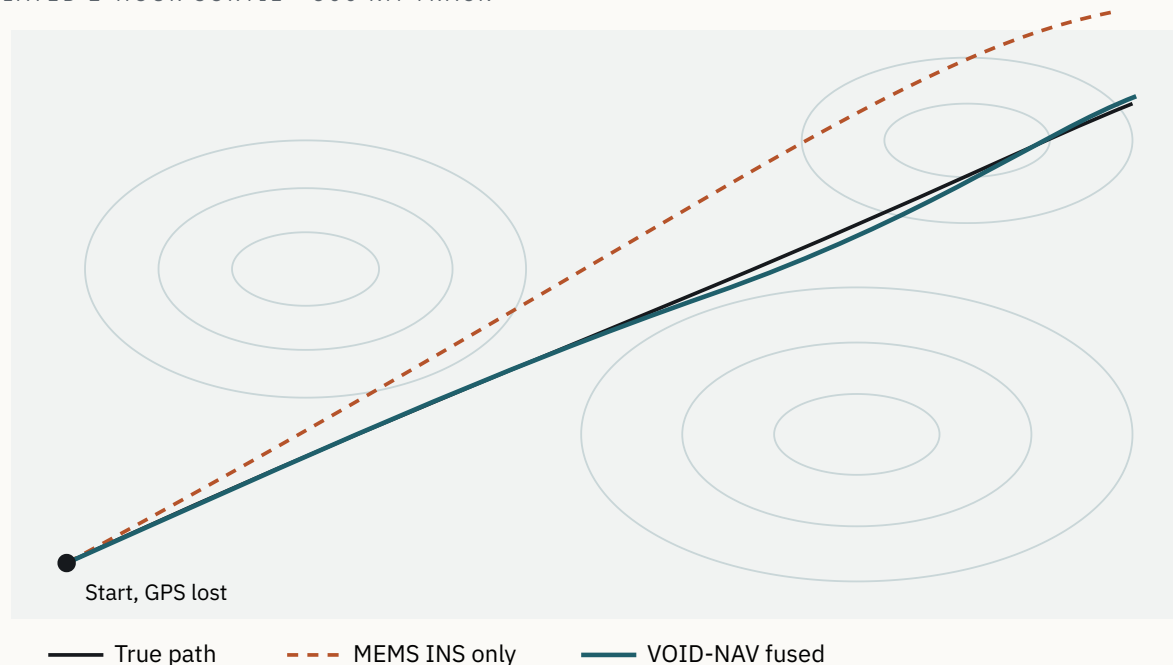
N - 04

Spoof detector for free

Running beside a GPS receiver, VOID-NAV flags a spoofing attack the moment GNSS and the magnetic fix disagree beyond their joint error bound.

In simulation over a real anomaly grid, drift stays bounded instead of growing.

SIMULATED 2-HOUR SORTIE • 300 KM TRACK



>10×

Target reduction in end-of-mission error versus MEMS INS alone.

<50 m

Target position fix over anomaly-rich terrain (airborne, low altitude).

~1 nT/ $\sqrt{\text{Hz}}$

Prototype sensitivity goal; sub-nT is reported for optimised NV ensembles.

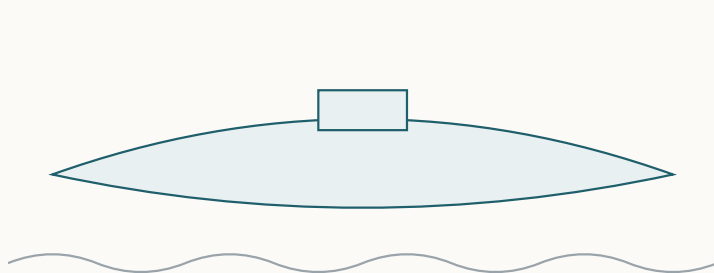
All values are design targets from simulation and published NV literature, to be validated on the prototype and in field trials.

Where VOID-NAV sits among alternatives.

APPROACH	JAM / SPOOF PROOF	LONG-TERM DRIFT	SIZE & POWER	COST	STEALTH
GNSS receiver	No	None, if signal present	Tiny	Low	Passive
MEMS / tactical INS	Yes	Grows with time	Small	Low–Med	Passive
Cold-atom interferometer	Yes	Very low	Large, vacuum + lasers	Very high	Passive
Terrain radar matching	Yes	Bounded	Medium	High	Emits, detectable
VOID-NAV (NV + INS)	Yes	Bounded	Compact, room-temp	Low	Passive

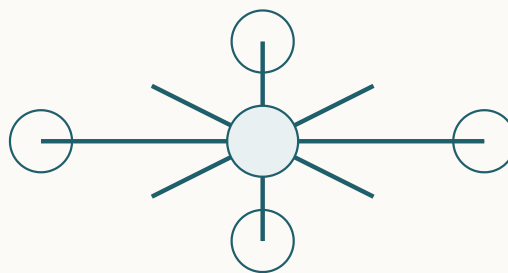
VOID-NAV complements cold-atom inertial sensors under the NQM roadmap: it delivers a deployable fix today, while atom-based INS matures.

One sensor core, three defence platforms.



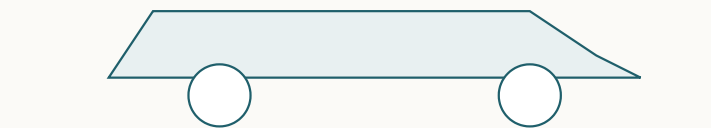
Submarines & UUVs

Weeks submerged with no satellite contact.
Magnetic fixes without surfacing preserve stealth.



UAVs & loitering systems

Resist jamming and spoofing over contested borders; low altitude maximises anomaly signal.



Border & coastal patrol

Reliable positioning for ground and maritime vehicles in high-altitude and jammed sectors.

Dual use: civil aviation resilience, maritime shipping, mineral exploration and geophysical survey, supporting AP's quantum valley ecosystem.

From a bench demo to a trial-ready unit.

HACKATHON · TRL 3



Bench prototype

Live ODMR spectrum
Field tracking vs magnet
Simulated map matching

+6 MONTHS · TRL 4



Vehicle-mounted unit

Road trials on surveyed
anomaly corridor

+12 MONTHS · TRL 5



Airborne trial

Drone flights with GNSS
as ground truth

+24 MONTHS · TRL 6



Ruggedised module

With DRDO / defence PSU
partners, MIL-STD tests

WHAT WE ASK FOR

Access to regional magnetic survey data · a trial range · mentorship from the NQM sensing hub · seed funding for TRL 4

SUMMARY

When the satellites go dark, the ground still knows where you are.

VOID-NAV: a passive, room-temperature quantum sensor for jam-proof navigation,
built in India and demonstrable today.

Team VOID

THANK YOU · QUESTIONS WELCOME

UC-085 · QUANTUM SENSING · DEFENCE